

Class 10 Science - Chapter 3

Metals and Non-metals (Hinglish Notes)

PART A - CORE BASICS (MISS MAT KARNA)

Class 7-8-9 ki yeh cheezein pehle pakki karo, warna Ch3 samajh nahi aayega.

1. Element, Metal, Non-metal (Class 8 base)

Element = ek hi type ke atoms se bani cheez (Fe, Cu, O, S).

Metal = aam taur par chamakdaar, kathor, bijli/garmi ka conductor.

Jaise: Iron (Fe), Copper (Cu), Aluminium (Al), Gold (Au), Sodium (Na).

Non-metal = aksar dull, bhurbhura/gas, bijli ka kuchalak (insulator).

Jaise: Oxygen (O), Carbon (C), Sulphur (S), Nitrogen (N), Chlorine (Cl).

Periodic table me: left+center = metals, right side = non-metals.

2. Atom, Ion, Valency, Electron (Class 9 base)

Atom ke andar: proton (+), neutron (0), electron (-).

Ion = charge wala atom. Cation = +ve (electron khoya, metals banate),

Anion = -ve (electron liya, non-metals banate).

Valency = combine karne ki capacity. Na=1, Mg=2, Al=3, O=2, Cl=1.

Metal electron DETA hai (cation), non-metal electron LETA hai (anion).

3. Ionic vs Covalent (idea level)

Ionic bond = electron ka TRANSFER (metal -> non-metal). Eg: NaCl.

Covalent bond = electron ka SHARING (non-metal + non-metal). Eg: H₂O, CH₄.

4. Common Symbols + Reaction bhasha

Symbols: Na, K, Ca, Mg, Al, Zn, Fe, Pb, Cu, Ag, Au, Hg(mercury).

Reactant (baayi) ----> Product (daayi). States (s)(l)(g)(aq).

Oxide = element + oxygen (jaise MgO, CO₂). Equation balance honi chahiye.

5. Oxidation/Reduction (Ch1 se link)

Oxidation = oxygen judna ya electron khona.

Reduction = oxygen hatna ya electron paana.

Metal aasani se oxidise hote (electron dete) - isiliye reactive.

-- END OF CORE BASICS (RED) PAGE --

PART B - CHAPTER KA MAIN CONTENT

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[GREEN = board exam me baar-baar poocha jaata hai]

1. Physical Properties

METALS ke physical properties:

- Lustre (chamak) hoti hai.
- Malleable: peet kar sheet (chaadar) banayi ja sakti (Au, Ag best).
- Ductile: kheech kar taar banaya ja sakta (Au sabse ductile).
- Sonorous: thokne par awaaz (ghanti) deti.
- Heat + electricity ke achhe conductor (silver + copper best).
- High melting point (zyadatar). Solid (except Mercury - liquid metal).

NON-METALS: dull, non-malleable (bhurbhura), insulator, kam melting point.

Exception: Graphite (carbon) bijli conduct karta; Iodine me lustre hota;
Diamond sabse kathor + high melting point (carbon ka roop).

Apvaad (exceptions) yaad rakho - exam me aate:

- Mercury (Hg): metal hokar bhi LIQUID.
- Sodium/Potassium: itne narm ki chaaku se kat jaate.
- Graphite: non-metal hokar bhi conductor.

2. Chemical Properties of Metals [TOP EXAM AREA]

(A) Oxygen ke saath - Metal oxide (basic)

Metal + O₂ ----> Metal oxide (yeh BASIC hote)



Amphoteric oxides: kuch metal oxides ACID + BASE dono se react karte.

Eg: Al₂O₃ aur ZnO (yaad rakho: "Al aur Zn = amphoteric").

Reactivity alag: Na/K thandi hawa me bhi aag pakad lete (kerosene me rakhte);

Mg jalta; Zn/Fe garam karne par; Au/Ag react hi nahi karte.

(B) Paani ke saath

Metal + Water ----> Metal hydroxide/oxide + Hydrogen

Na + thanda paani -> NaOH + H₂ (bahut tezi, aag pakad sakti).

Mg + garam paani -> Mg(OH)₂ + H₂.

Fe + steam (bhaap) -> Fe₃O₄ + H₂.

Cu, Ag, Au paani se react NAHI karte (least reactive).

(C) Acid ke saath

Metal + Dilute acid ----> Salt + Hydrogen gas



Copper, dilute HCl se react NAHI karta (H ke neeche reactive series me).
Aqua regia (3:1 conc.HCl : conc.HNO₃) - Gold/Platinum ko bhi ghol deta.

3. Reactivity Series [VERY IMPORTANT]

Ghatti hui reactivity ka kram:



Memory: "Kya Naani Ka Magaz Aluminium Zinc Faad Lega Pet Hydrogen
Copper Hilakar Silver Gold".

Upar wala metal, neeche wale ke salt se usko HATA (displace) deta hai.



4. Ionic Compounds (Metal + Non-metal) [IMPORTANT]

Metal electron deta, non-metal leta -> ionic (electrovalent) bond.

Properties of ionic compounds:

- Solid + kathor (crystalline).
- High melting + boiling point.
- Paani me ghulte (zyadatar); kerosene/petrol me nahi.
- Pighli (molten) ya ghuli (aqueous) halat me bijli conduct karte (kyunki ions free ho jaate); thos halat me NAHI.

5. Metallurgy - Metals ko nikalna (Extraction) [TOP EXAM AREA]

Mineral = element wala natural padarth. Ore = jis mineral se sasta+aasan metal nikle. Gangue = ore me mili mitti/impurity.

Reactivity ke hisaab se extraction:

- (1) Low reactive (Au, Ag, Pt, Cu) - prakriti me FREE/aise hi milte.
- (2) Medium (Zn, Fe, Pb) - oxide ko CARBON se reduce karke.
- (3) High (K, Na, Ca, Mg, Al) - ELECTROLYSIS (bijli) se reduce karte.

Steps:

Enrichment (gangue hatana) -> Roasting / Calcination -> Reduction -> Refining.

Roasting: sulphide ore ko hawa me garam (ZnS -> ZnO). $2ZnS + 3O_2 \rightarrow 2ZnO + 2SO_2$

Calcination: carbonate ore ko hawa ke bina garam (ZnCO₃ -> ZnO + CO₂).

Reduction: ZnO + C -> Zn + CO (oxide ko metal banaya).

Thermit reaction: $Fe_2O_3 + 2Al \rightarrow 2Fe + Al_2O_3$ + bahut heat
(rail patri jodne me - pighla loha milta).

Refining: Electrolytic refining (impure metal anode, shudh cathode par jamta).

6. Corrosion aur Bachaav (Ch1 se link) [IMPORTANT]

Corrosion = metal ka surface se dheere kharab hona (air+moisture).

Iron par RUST (Fe₂O₃.xH₂O, bhura), Silver kala (Ag₂S), Copper hara.

Rusting rokne ke tareeke: painting, oiling, greasing, galvanisation

(Zn coating), chrome plating, alloy banana, tinning.

Galvanisation = iron par zinc ki layer (zinc pehle corrode hota, iron bacha).

7. Alloys (Mishra-dhatu) [IMPORTANT]

Alloy = do ya zyada metals (ya metal+non-metal) ka homogeneous mixture.

Banane ka reason: zyada strong, rust-resistant, ya properties improve.

Examples (RATTA - exam me direct):

- Steel = Iron + Carbon.
- Stainless steel = Iron + Chromium + Nickel (rust nahi).
- Brass (Peetal) = Copper + Zinc.
- Bronze (Kansa) = Copper + Tin.
- Solder = Lead + Tin (kam melting point - soldering me).
- Amalgam = koi metal + Mercury.

24-carat gold = pure (narm); jewellery me 22-carat (gold + copper/silver).

8. Quick Revision - One Liners

- Metal: electron deta, basic oxide, malleable+ductile, conductor.
- Non-metal: electron leta, acidic oxide, brittle, insulator.
- Reactivity: K Na Ca Mg Al Zn Fe Pb H Cu Hg Ag Au.
- Amphoteric oxide: Al_2O_3 , ZnO .
- Thermit: $\text{Fe}_2\text{O}_3 + \text{Al} \rightarrow \text{Fe}$ (rail welding).

PART C - SOLVED EXAMPLES (HARDEST -> EASIEST)

Example 1 (HARDEST) - Extraction with reactivity logic

Q: Zinc, zinc blende (ZnS) se kaise nikalte hain? Saare steps + equations.

Solution:

- Step 1 Roasting: ZnS ko hawa me garam $\rightarrow 2\text{ZnS} + 3\text{O}_2 \rightarrow 2\text{ZnO} + 2\text{SO}_2$.
- Step 2 Reduction: ZnO ko carbon se reduce $\rightarrow \text{ZnO} + \text{C} \rightarrow \text{Zn} + \text{CO}$.
- Step 3 Refining: electrolytic refining se shudh zinc.

Zinc medium-reactive hai isliye carbon reduction kaam karta (electrolysis ki zaroorat nahi, jaise Na/Al me hoti).

Example 2 - Reactivity + displacement decide karna

Q: Fe ki keel (nail) ko CuSO_4 ke neele ghol me dubaayein to kya hoga? Kyun?

Solution:

- Iron, copper se zyada reactive hai (series me upar).
- $\text{Fe} + \text{CuSO}_4 \rightarrow \text{FeSO}_4 + \text{Cu}$. Keel par bhura copper jamta, ghol ka neela rang halka (green-ish FeSO_4) ho jaata. Yeh displacement reaction.

Example 3 - Amphoteric oxide

Q: Aluminium oxide ko amphoteric kyun kehte? Equations do.

Solution:

- Amphoteric = acid + base dono se react kare.
- Acid ke saath: $\text{Al}_2\text{O}_3 + 6\text{HCl} \rightarrow 2\text{AlCl}_3 + 3\text{H}_2\text{O}$.
- Base ke saath: $\text{Al}_2\text{O}_3 + 2\text{NaOH} \rightarrow 2\text{NaAlO}_2 + \text{H}_2\text{O}$.

Example 4 - Ionic compound property

Q: NaCl ka melting point zyada kyun? Solid me bijli conduct kyun nahi karta?

Solution:

- Strong ionic bond (Na^+ aur Cl^- ke beech zabardast attraction) $\rightarrow$ todne ko bahut energy chahiye $\rightarrow$ high melting point.
- Solid me ions fix (move nahi kar sakte) $\rightarrow$ conduct nahi; molten/aqueous me ions free $\rightarrow$ tab conduct karta.

Example 5 (EASIEST) - Metal vs Non-metal pehchaan

Q: Ek padarth chamakdaar, taar me khinch jaata, bijli conduct karta - metal ya non-metal?

Solution:

- Lustre + ductile + conductor -> yeh METAL hai.

PART D - SELF TEST (EASY -> HARD)

EASY (1 mark)

- Q1. Malleable aur ductile ka matlab batao.
- Q2. Sabse zyada aur sabse kam reactive metal kaunsa (series se)?
- Q3. Amphoteric oxide kya hai? 2 example.
- Q4. Steel aur brass kis-kis se bante hain?

MEDIUM (2-3 marks)

- Q5. Metal + acid, metal + water ke general reaction + 1-1 example.
- Q6. Roasting aur calcination me antar (equation sahit).
- Q7. Ionic compounds ke 3 properties likho.
- Q8. Galvanisation kya hai aur yeh iron ko kaise bachata hai?

HARD (3-5 marks, board favourite)

- Q9. ZnS se zinc nikalne ke saare steps + equations.
- Q10. Reactivity series likho aur batao kyun Cu, FeSO₄ ko displace nahi karta.
- Q11. Thermit reaction kya hai? Equation + use.
- Q12. Corrosion kya hai? 3 metals ke corrosion + 3 bachaav ke tareeke.

ANSWER HINTS (PART D)

- A1. Malleable: sheet ban sake; Ductile: taar ban sake.
- A2. Sabse reactive K (potassium); sabse kam Au (gold).
- A3. Acid+base dono se react -> Al₂O₃, ZnO.
- A4. Steel = Fe + C; Brass = Cu + Zn.
- A5. Metal+acid->salt+H₂ (Zn+H₂SO₄); metal+water->hydroxide+H₂ (Na+H₂O).
- A6. Roasting = sulphide hawa me garam (ZnS->ZnO); Calcination = carbonate bina hawa garam (ZnCO₃->ZnO+CO₂).
- A7. High m.p., solid+kathor, molten/aqueous me conduct, paani me ghulte.
- A8. Iron par zinc layer; zinc pehle corrode hokar iron ko bachata.
- A9. Roasting (2ZnS+3O₂->2ZnO+2SO₂), reduction (ZnO+C->Zn+CO), refining.
- A10. K..Na..Au; Cu, Fe se kam reactive isliye FeSO₄ se Fe ko nahi hata sakta.
- A11. Fe₂O₃+2Al->2Fe+Al₂O₃+heat; rail patri welding.
- A12. Air+moisture se kharab; iron rust, silver Ag₂S, copper hara; painting/oiling/galvanisation.

"Padhai tabhi pakki jab khud likh ke test do." - All the best, Krishna!

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